

Data Science – Introduction

What is Data Science?

Data Science is an interdisciplinary field that uses scientific methods, algorithms, and systems to extract knowledge and insights from data.

Combines:

- Statistics
- Computer Science
- Domain Knowledge

Key Goals

- Extract meaningful insights
- Support decision-making
- Predict future trends

```
In [1]: # Simple example: average marks
marks = [85, 78, 92, 88, 76]
average = sum(marks) / len(marks)
print("Average Marks:", average)
```

Average Marks: 83.8

Structured Data

Organized in rows & columns. Easy to analyze.

Student_ID	Name	Marks	Grade
101	Ravi	85	A
102	Sita	78	B
103	Kumar	92	A

```
In [2]: import pandas as pd

data = {
    "Student_ID": [101, 102, 103],
    "Name": ["Ravi", "Sita", "Kumar"],
    "Marks": [85, 78, 92],
    "Grade": ["A", "B", "A"]
}
```

```
df = pd.DataFrame(data)
print(df)
```

	Student_ID	Name	Marks	Grade
0	101	Ravi	85	A
1	102	Sita	78	B
2	103	Kumar	92	A

Unstructured Data

No fixed format. Examples: text, images, videos

```
In [3]: text = "Data Science is an evolving field"
words = text.split()
print(text)
print(words)
print("Word Count:", len(words))
```

```
Data Science is an evolving field
['Data', 'Science', 'is', 'an', 'evolving', 'field']
Word Count: 6
```

Semi-Structured Data

Definition

Semi-structured data is data that does not follow a strict tabular format but contains some level of organization using:

- Keys
- Tags
- Markers

Examples

- JSON
- XML
- HTML

Example (JSON)

```
{
  "Student_ID": 101,
  "Name": "Ravi",
  "Marks": 85,
  "Grade": "A"
}
```

 Not in table form

 Organized using key-value pairs

```
In [ ]: # Semi-structured data example using JSON

import json
```

```

json_data = '''
{
    "Student_ID": 101,
    "Name": "Ravi",
    "Marks": 85,
    "Grade": "A"
}
'''

data = json.loads(json_data)

print("Complete Data:", data)
print("Student Name:", data["Name"])
print("Marks:", data["Marks"])

```

Summary

- Structured → Tables
- Semi-structured → JSON/XML
- Unstructured → Text, images

```

In [1]: # Identify data types
import pandas as pd

data1 = {"Name": "Ravi", "Marks": 85}
data2 = "This is a text document"
data3 = pd.DataFrame({"Marks": [85, 90]})

print(data1)
print(data2)
print(data3)

```

```

{'Name': 'Ravi', 'Marks': 85}
This is a text document
   Marks
0      85
1      90

```

In []: